Elementary Analysis

Section 26: Differentiation and Integration of Power Series

Exercises 26.1–26.8, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

26 Differentiation and integration of power series **2**

26.1 Exercise 26.1: Term-by-term integration at a positive input 2

26.2 Exercise 26.2: A differentiated geometric series and three numerical sums 2

26.3 Exercise 26.3: One more derivative gives squared indices 3

26.4 Exercise 26.4: Integrating the Gaussian power series 4

26.5 Exercise 26.5: A function equal to its own derivative, without naming it 4

26.6 Exercise 26.6: A trigonometric identity derived without trigonometry 5

26.7 Exercise 26.7: Why absolute value has no power-series representation at zero 5

26.8 Exercise 26.8: The arctangent series and its endpoints 6

26 Differentiation and integration of power series

The permission comes before the calculation. Inside the radius of convergence of a power series, we may differentiate term by term, and we may integrate term by term from zero to any interior point. Uniform convergence on each smaller closed interval is the reason integration works. Endpoints require a separate argument: convergence inside an interval does not by itself settle either endpoint. We use Abel's theorem only after checking that the numerical endpoint series converges.

26.1 Exercise 26.1: Term-by-term integration at a positive input

What we must show. If $f(t) = \sum_{k=0}^{\infty} a_k t^k$ has radius $R > 0$, prove $\int_0^x f(t)\,dt = \sum_{k=0}^{\infty} a_k x^{k+1}/(k+1)$ when $0 < x < R$.

Work on an interval lying strictly inside the radius.

Fix such an x, and put $s_n(t) = \sum_{k=0}^{n} a_k t^k$. The power series converges uniformly on $[-x, x]$ by Theorem 26.1, and therefore also on $[0, x]$. Each s_n is a continuous polynomial.

Pass the limit through the integral using the established theorem.

Theorem 25.2 yields

$$\int_0^x f(t)\,dt = \lim_{n\to\infty} \int_0^x s_n(t)\,dt.$$

For a finite sum, linearity and the power rule give

$$\int_0^x s_n(t)\,dt = \sum_{k=0}^{n} a_k \int_0^x t^k\,dt = \sum_{k=0}^{n} \frac{a_k}{k+1} x^{k+1}.$$

Taking the limit of these finite sums gives precisely the claimed infinite series.

Conclusion. The term-by-term integration formula holds for every positive $x < R$. The strict inequality $x < R$ is what secured the uniform convergence used in the proof.

26.2 Exercise 26.2: A differentiated geometric series and three numerical sums

What we must show. Derive $\sum_{n=1}^{\infty} nx^n$ and use it to evaluate the requested sums.

(a) Differentiate first, then restore the missing power of x.

For $|x| < 1$, the geometric series is $\sum_{n=0}^{\infty} x^n = 1/(1-x)$. Its radius is 1, so Theorem 26.5 allows differentiation at every such x:

$$\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

Multiplying by x gives

$$\boxed{\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}} \qquad (|x| < 1).$$

(b) Substitute $x = 1/2$.

The value lies strictly inside the radius, so the formula gives

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1/2}{(1-1/2)^2} = \frac{1/2}{1/4} = 2.$$

(c) Substitute the two signed inputs separately.

At $x = 1/3$,

$$\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{1/3}{(2/3)^2} = \frac{3}{4}.$$

At $x = -1/3$,

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{3^n} = \frac{-1/3}{(1+1/3)^2} = \frac{-1/3}{16/9} = -\frac{3}{16}.$$

Conclusion. The three numerical sums are 2, 3/4, and $-3/16$.

26.3 Exercise 26.3: One more derivative gives squared indices

What we must show. Find $\sum_{n=1}^{\infty} n^2 x^n$ and evaluate it at 1/2 and 1/3.

(a) Differentiate the identity from Exercise 26.2.

For $|x| < 1$, term-by-term differentiation is legitimate, because differentiation preserves the radius of a power series. Hence

$$\sum_{n=1}^{\infty} n^2 x^{n-1} = \frac{d}{dx}\left[\frac{x}{(1-x)^2}\right] = \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^3} = \frac{1+x}{(1-x)^3}.$$

Multiplication by x gives

$$\boxed{\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}} \qquad (|x| < 1).$$

(b) Evaluate the two substitutions.

Both inputs are strictly inside the radius. At $x = 1/2$ the numerator is $(1/2)(3/2) = 3/4$ and the denominator is $(1/2)^3 = 1/8$, so

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \frac{3/4}{1/8} = 6.$$

At $x = 1/3$ the numerator is $4/9$ and the denominator is $8/27$, so

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} = \frac{4/9}{8/27} = \frac{3}{2}.$$

Conclusion. The general formula is $x(1+x)/(1-x)^3$, and the requested values are 6 and 3/2.

26.4 Exercise 26.4: Integrating the Gaussian power series

What we must show. Express e^{-x^2} and $F(x) = \int_0^x e^{-t^2}\,dt$ as power series.

(a) Substitute into the exponential series.

The exponential identity, with its constant term included, is

$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} \qquad (u \in \mathbb{R}).$$

Taking $u = -x^2$ gives

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots .$$

This substitution is valid for every real x because the original exponential series converges for every real input.

(b) Verify the radius before integrating.

For fixed x, the ratio of successive absolute terms is $x^2/(n+1) \to 0$, so this power series has infinite radius. Theorem 26.4 therefore applies between zero and every real x, whether positive or negative. It gives

$$\boxed{F(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!}} = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots .$$

The lower limit contributes zero for each monomial, so no integration constant has been omitted.

Conclusion. The two displayed expansions hold for all $x \in \mathbb{R}$.

26.5 Exercise 26.5: A function equal to its own derivative, without naming it

What we must show. For $f(x) = \sum_{n=0}^{\infty} x^n/n!$, prove $f' = f$ directly from the series.

Check that term-by-term differentiation is allowed everywhere.

For each fixed x, the ratio of successive absolute terms is $|x|/(n+1) \to 0$. Thus the radius is infinite, and Theorem 26.5 allows term-by-term differentiation at every real input.

Differentiate and carefully change the index.

The constant term differentiates to zero, and for $n \geq 1$, $n/n! = 1/(n-1)!$. Therefore

$$f'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{j=0}^{\infty} \frac{x^j}{j!} = f(x).$$

In the last step we set $j = n - 1$, so the first new index is zero.

Conclusion. $f' = f$ on $\mathbb{R}$. No identification of f with the exponential function was used.

26.6 Exercise 26.6: A trigonometric identity derived without trigonometry

What we must show. For

$$s(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \qquad c(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

prove the requested derivative relations and $s^2 + c^2 = 1$.

(a) Differentiate each series with its radius justified.

For fixed x, ratios of successive absolute terms tend to zero in both series, since their denominators acquire two factorial factors while their numerators acquire only a factor x^2. Thus both radii are infinite. Theorem 26.5 gives

$$s'(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = c(x).$$

For c, the $n = 0$ term differentiates to zero, so

$$\begin{aligned} c'(x) &= \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!} \\ &= -\sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{(2j+1)!} = -s(x). \end{aligned}$$

Here $j = n - 1$, and $(-1)^{j+1} = -(-1)^j$ explains the minus sign.

(b) Differentiate the sum of squares.

The product rule gives

$$(s^2 + c^2)' = 2ss' + 2cc' = 2sc + 2c(-s) = 0.$$

(c) Determine the constant from one input.

A differentiable function with derivative zero on an interval is constant there, by the Mean Value theorem. Apply this on the interval joining zero to any real x. Thus $s(x)^2 + c(x)^2 = s(0)^2 + c(0)^2$. The defining series give $s(0) = 0$ and $c(0) = 1$, so that constant is 1.

Conclusion. $s' = c$, $c' = -s$, and $s(x)^2 + c(x)^2 = 1$ for every real x, proved using the series alone.

26.7 Exercise 26.7: Why absolute value has no power-series representation at zero

What we must show. Decide whether $|x| = \sum_{n=0}^{\infty} a_n x^n$ can hold for all real x.

A power series with positive radius is differentiable at its center.

If such a representation held for every real input, its radius would be infinite, in particular positive. Theorem 26.5 would make its sum differentiable at zero.

Calculate the two difference quotients of absolute value.

Since $|0| = 0$,

$$\frac{|h| - |0|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

The right-hand and left-hand limits differ, so $|x|$ is not differentiable at zero. This contradicts the consequence of a power-series representation.

Conclusion. No such power series exists, even on any two-sided neighborhood of zero. On the separate half-lines the formulas are $|x| = x$ and $|x| = -x$, but these do not combine into one power series about zero. Polynomial approximation, discussed next, is a different and weaker requirement.

26.8 Exercise 26.8: The arctangent series and its endpoints

What we must show. Derive the arctangent series inside $(-1, 1)$ and settle both endpoints.

(a) Substitute into the geometric series.

For $|x| < 1$, put $y = -x^2$, so $|y| < 1$. Then

$$\sum_{n=0}^{\infty} (-1)^n x^{2n} = \sum_{n=0}^{\infty} y^n = \frac{1}{1-y} = \frac{1}{1+x^2}.$$

(b) Integrate on a strictly interior interval.

The power series has radius 1, so Theorem 26.4 gives, for $|x| < 1$,

$$\int_0^x \frac{dt}{1+t^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}.$$

The integral on the left is $\arctan x - \arctan 0 = \arctan x$. Therefore

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \qquad (-1 < x < 1).$$

(c) Establish convergence at 1 before using Abel's theorem.

At $x = 1$ the numerical series is $\sum_{n=0}^{\infty} (-1)^n/(2n+1)$. Its positive term sizes decrease to zero, so the alternating-series test proves convergence. Abel's theorem then identifies its sum with the limit of the interior sum as $x \uparrow 1$. Since arctangent is continuous,

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \lim_{x \uparrow 1} \arctan x = \frac{\pi}{4}.$$

Multiplication by 4 gives the familiar formula

$$\boxed{\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)}.$$

(d) Evaluate at the negative endpoint with its sign intact.

Every power in the series is odd. Substitution $x = -1$ therefore makes the series the negative of the convergent series in part (c). Its sum is $-\pi/4 = \arctan(-1)$. It converges conditionally at both endpoints: the series of absolute values $\sum 1/(2n+1)$ diverges by comparison with the harmonic series, since $1/(2n+1) \geq 1/(2n+2)$.

Conclusion. The arctangent expansion holds on $[-1, 1]$, with endpoint values $\pi/4$ and $-\pi/4$ justified separately, rather than by interior convergence alone.